CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 13: Pythagoras Theorem

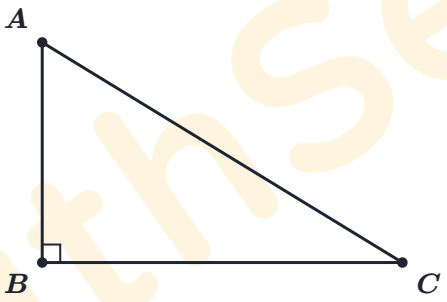
Topic-wise short notes with solved examples

Pythagoras theorem and its converse, Pythagorean triplets, and their use in finding lengths in triangles, rectangles, rhombuses and everyday situations.

1 The theorem

- In a **right-angled triangle**, the square on the **hypotenuse** equals the sum of the squares on the other two sides.
- If $\angle B = 90^\circ$ in triangle ABC, then

$$AC^2 = AB^2 + BC^2$$



Angle B is 90°, so AC² = AB² + BC²

- The hypotenuse is the side opposite the right angle. It is the longest side.
- **A proof by area**: draw BD perpendicular to AC. Triangles ABD and ACB have the same angles, which gives $AB^2 = AD \times AC$. In the same way $BC^2 = DC \times AC$. Adding: $AB^2 + BC^2 = AC(AD + DC) = AC^2$.

Example 1.1 In triangle ABC, $\angle B = 90^\circ$, $AB = 9$ cm and $BC = 12$ cm. Find AC.

SOLUTION

$AC^2 = 81 + 144 = 225$

15 cm

Example 1.2 In triangle PQR, $\angle Q = 90^\circ$, $PR = 25$ cm and $PQ = 7$ cm. Find QR.

SOLUTION

$QR^2 = 625 - 49 = 576$

24 cm

2 Converse

- If the square on one side of a triangle equals the sum of the squares on the other two sides, then the angle **opposite the first side** is a right angle.
- **To test a triangle**: square the longest side and compare with the sum of the squares of the other two.

Comparison (c is the longest side)	Triangle
$c^2 = a^2 + b^2$	Right-angled
$c^2 < a^2 + b^2$	Acute-angled
$c^2 > a^2 + b^2$	Obtuse-angled

Example 2.1 Is a triangle with sides 8 cm, 15 cm and 17 cm right-angled?

SOLUTION

$17^2 = 289$ and $8^2 + 15^2 = 64 + 225 = 289$

Yes; the right angle is opposite the 17 cm side

Example 2.2 What kind of triangle has sides 5 cm, 6 cm and 9 cm?

SOLUTION

$81 > 25 + 36 = 61$

Obtuse-angled

3 Pythagorean triplets

- Three natural numbers a, b, c with $a^2 + b^2 = c^2$ form a **Pythagorean triplet**.
- Common triplets: $(3, 4, 5)$, $(5, 12, 13)$, $(8, 15, 17)$, $(7, 24, 25)$, $(20, 21, 29)$, $(9, 40, 41)$.
- Any multiple of a triplet is a triplet: $(6, 8, 10)$, $(9, 12, 15)$, $(10, 24, 26)$.
- For any natural number $m > 1$: $(2m, m^2 - 1, m^2 + 1)$ is a triplet.

Example 3.1 Write a Pythagorean triplet one of whose members is 12.

SOLUTION

Take $2m = 12$, so $m = 6$: $m^2 - 1 = 35$ and $m^2 + 1 = 37$.

$(12, 35, 37)$

4 Standard applications

- **Diagonal of a rectangle**: $d^2 = l^2 + b^2$. **Diagonal of a square** of side a: $a\sqrt{2}$.
- **Rhombus**: the diagonals bisect each other at right angles, so $\text{side}^2 = \left(\frac{d_1}{2}\right)^2 + \left(\frac{d_2}{2}\right)^2$, that is $4 \times \text{side}^2 = d_1^2 + d_2^2$.

- **Isosceles triangle**: the altitude to the base bisects the base.
- **Equilateral triangle** of side a: altitude $= \frac{\sqrt{3}}{2}a$ and area $= \frac{\sqrt{3}}{4}a^2$.
- **Altitude on the hypotenuse**: if the legs are a, b and the hypotenuse is c, the altitude is $\frac{ab}{c}$.

Example 4.1 A rectangle is 24 cm long and its diagonal is 26 cm. Find its breadth.

SOLUTION

$b^2 = 676 - 576 = 100$

10 cm

Example 4.2 The diagonals of a rhombus are 16 cm and 30 cm. Find its perimeter.

SOLUTION

Side$^2 = 8^2 + 15^2 = 289$, so the side is 17 cm.

68 cm

Example 4.3 Find the altitude of an equilateral triangle of side 8 cm.

SOLUTION

$h^2 = 8^2 - 4^2 = 48$

$4\sqrt{3}$ cm, about 6.93 cm

Example 4.4 The legs of a right triangle are 6 cm and 8 cm. Find the altitude on the hypotenuse.

SOLUTION

Hypotenuse $= 10$ cm. Area $= \frac{1}{2} \times 6 \times 8 = \frac{1}{2} \times 10 \times h$.

4.8 cm

5 Problems from daily life

- Draw the figure, mark the right angle and name the three sides before using the theorem.

Example 5.1 A ladder 13 m long reaches a window 12 m above the ground. How far is the foot of the ladder from the wall?

SOLUTION

$d^2 = 169 - 144 = 25$

5 m

Example 5.2 Two poles of heights 6 m and 11 m stand 12 m apart on level ground. Find the distance between their tops.

SOLUTION

The tops differ in height by 5 m and are 12 m apart horizontally: $d^2 = 25 + 144$.

13 m

Example 5.3 A man goes 15 m due west and then 8 m due north. How far is he from the starting point?

SOLUTION

$d^2 = 225 + 64 = 289$

17 m

6 Results in triangles

- In triangle ABC with AD perpendicular to BC: $AB^2 - BD^2 = AC^2 - CD^2$ (both equal AD^2).
- If $\angle B = 90^\circ$ and D is the mid-point of BC: $AC^2 = AD^2 + 3CD^2$.
- In an equilateral triangle with altitude AD: $3AB^2 = 4AD^2$.

Example 6.1 In triangle ABC, AD is perpendicular to BC, $AB = 13$ cm, $BD = 5$ cm and $DC = 9$ cm. Find AC.

SOLUTION

$AD^2 = 169 - 25 = 144$. Then $AC^2 = 144 + 81 = 225$.

15 cm

Example 6.2 In triangle ABC, $\angle B = 90^\circ$ and D is the mid-point of BC. Prove that $AC^2 = AD^2 + 3CD^2$.

SOLUTION

$AC^2 = AB^2 + BC^2$ and $AD^2 = AB^2 + BD^2$. Subtracting: $AC^2 - AD^2 = BC^2 - BD^2$.

$BC = 2CD$ and $BD = CD$, so $BC^2 - BD^2 = 4CD^2 - CD^2$.

$AC^2 = AD^2 + 3CD^2$

7 Common mistakes

- Adding when a shorter side is required: use hypotenuse^2 minus the known side^2.
- Taking the wrong side as the hypotenuse. It is always opposite the right angle.
- Forgetting to take the square root at the end.
- Using the full diagonals of a rhombus instead of the half-diagonals.
- Testing the converse with a side that is not the longest.
- Using the theorem in a triangle that is not right-angled.

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